ARVIN BOZORGI

Glenelg, MD | (443)-430-5034 | bozorgi@gmail.com | arvinbozorgi.com | GitHub

Education

- *McDonogh High School,* Expected Graduation: 2027
- Dean's List 2024, '25, '26
- 2026 Computer Science Award recipient - Presented for outstanding achievement and interest in computer science
- 35 ACT

Experience

UMD Hackathon *February 2026*

- Competed in a Hackathon hosted by UMD, representing and recognized for my quick problem-solving and my ability to collaborate with a team.

FRC Team, *Programmer for McDonogh School* *January 2024 - Present*

- Programmer for the school's FRC team, developing robot control software in Java using WPILib and other FRC libraries for competition robots.

Glaucoma Classification Research, *Intern* *June 2025 - Present*

- Drafting research paper, "Impact of Different Data Augmentation Techniques for Glaucoma Detection Using a SWIN Transformer," with Dr. Javidian for publication.
- Annotated 1,500+ retinal images in Roboflow and trained a model to isolate optic nerve regions for focused glaucoma detection.
- Building PyTorch classification models and testing augmentation strategies to address dataset imbalance while tracking metrics in Weights & Biases.

Conceptbytes Team, *Developer [Demonstration]* *August 2024 - Present*

- Collaborate with a well-known influencer and engineer to develop an innovative AI project featured on a professional platform, gaining significant attention.
- Develop computer vision systems using Python, MediaPipe, and OpenCV.

A Byte For All, *Developer & Project Branch Director* *June 2025 - Apr 2026*

- Lead a team of student developers through community projects, such as web development.
- Create websites (HTML, CSS, JS) for nonprofits, businesses, and school clubs.

High School STEM Center, *Tutor* *August 2025 - Present*

- Selected through a competitive interview based on STEM coursework and soft skills to tutor students during school hours.
- Guide students through problems without revealing solutions.

Projects

Computer Vision Chess Glasses *March 2025*
- Trained a model to detect chessboard corners and warp live footage for analysis.
- Tracked piece movement using binary thresholding and canny edge detection.
- Streamed video from Meta Raybans and displayed moves in real time with PyGame.

Soccer Tic-tac-toe *[Demonstration]* *December 2024*
- Hand-annotated 4,000+ images to train a computer vision model that tracks a soccer ball in real time. Used this to project a PyGame tic-tac-toe board onto a wall and detected ball impact location to register moves.
- Created an alternative (projector + laptop + webcam) to $5,000 commercial systems.

Warfields II Home Owners Association Website *[Website]* *June 2025*
- Developed a modern and accessible HOA website based on board member feedback.

Jarvis (Hand-Tracking Projected UI), *[Demonstration]* *August 2024 - Present*
- Built a menu system and two arcade apps (Space Invaders, Brick Breaker) for Jarvis, an open-source Iron Man–style projected UI using OpenCV, MediaPipe, and PyGame.
- Sold apps to the project founder for $200; featured on his Instagram page.
- Joined the official GitHub developer team and remain an ongoing contributor.

Independent Author, *Terror Tales* *May 2023*
- Wrote and independently published a book containing five original short stories.
- Managed the full writing, editing, and Amazon (KDP) publication process.

Leadership

Competitive Programming Club, *President & Founder* *May 2025 - Present*
- Lead discussions on competition strategy and coordinate teams to represent our school in regional programming contests.

Founder of Machine Learning Club, *President & Founder* *May 2025 - Present*
- Taught neural network programming and machine learning applications.
- Create presentations and Python code walkthroughs for beginner-level models.

Founder of 3D Design Club *August 2024 - May 2025*
- Founded and led a CAD and 3D printing club, teaching Fusion 360 and prototyping.

Technical Skills:
- **Programming**: Python, Java, C, C#, C++, HTML, JavaScript, CSS, Lua, SQL
- **AI & Machine Learning**: TensorFlow, PyTorch, OpenCV, MediaPipe, Weights & Biases, Roboflow, Neural networks, facial recognition, hand tracking
- **3D Design & Prototyping**: Fusion 360, Blender modeling, 3D Printing
- **Certificates:** PCEP – Certified Entry-Level Python Programmer, HarvardX CS50